

FILTER DESIGN TEST CASE

Duplexer Module - Band 5 (850 MHz) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0034
Revision:	Rev C.4
Author:	Ana Reyes
Reviewer:	Dr. Olaf Bergstrom
Design Center:	Newbury Park Design Center
Created Date:	2025-03-28
Last Modified:	2025-04-03
Status:	PASS
Priority:	P4 - Low

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting Band 5 (850 MHz) for Millimeter-Wave Beamforming. The filter is designed on AlN on Si substrate with a target center frequency of 3221.1 MHz and bandwidth of 201.5 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: Band 5 (850 MHz)
- Application: Millimeter-Wave Beamforming
- Substrate: AlN on Si
- Package: Flip-Chip BGA
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Duplexer Module design targeting Band 5 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3221.1 MHz
Bandwidth (3 dB):	201.5 MHz
Insertion Loss Target:	<= 3.3 dB
Return Loss Target:	>= 20.0 dB
Out-of-Band Rejection:	>= 27.0 dB
Isolation (Duplexer):	>= 23.0 dB
Q Factor:	4555
Temperature Coefficient:	-28.9 ppm/°C
Die Size:	0.8 x 1.4 mm
Wafer Size:	6-inch
Number of Resonators:	7
Electrode Material:	Mo/Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Duplexer Module template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3221.1 MHz, BW 201.5 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9663 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.3 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 302 MHz offset >= 27.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 11.6 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2025-04-03
Devices Tested: 27
Insertion Loss (meas): 2.98 dB
Return Loss (meas): 21.9 dB
Rejection (meas): 29.6 dB
Isolation (meas): 28.3 dB
Q Factor (meas): 4555
Group Delay Variation: 6.8 ns
Power Handling: +34 dBm
Die Yield: 96.0%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated excellent performance for Band 5 (850 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author:	Ana Reyes	Date:	2025-04-03
Reviewer:	Dr. Olaf Bergstrom	Date:	2025-04-09
Approver:	Dr. Hans Weber	Date:	2025-04-11

END OF DOCUMENT - FD-TC-0034